CAMERA MODULE TECHNICAL DATASHEET

Model: [TS2M3189] **Revision:** [V1.0]

1. Product Overview

PRODUCT IMAGE	
	Model: **TS2M3189** Image Sensor: **HM2131** Resolution: **2MP** Interface: **USB 2.0** Frame Rate: **1080P @ 30fps (MJPG)** Module Size: **30 × 58 × 16 mm**

Product Description

The TS2M3189 is a USB 2.0 stereo camera module featuring HM2131 image sensors and a compact 30 × 58 × 16 mm mechanical design. It supports 1920 × 1080 imaging at 30fps in MJPG format, with YUV and MJPG output formats specified by the supplier. Its optical configuration includes 650 nm and 850 nm wavelength specifications, making it a candidate for applications requiring visible-light and infrared-related imaging configurations. Final application suitability should be confirmed against the optical and system integration requirements.

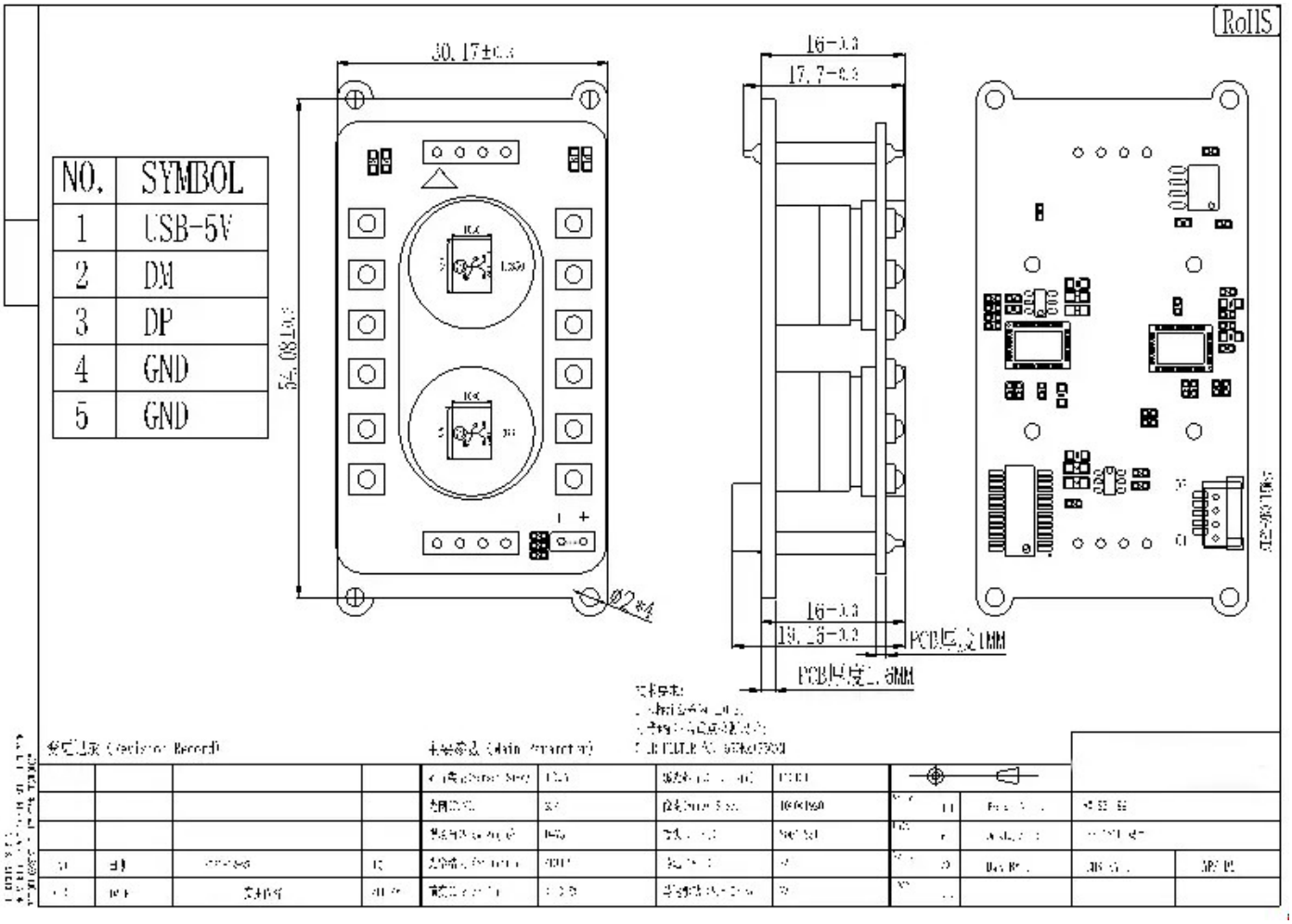
2. Key Features

- 2MP imaging with 1920 × 1080 resolution
- 1080P @ 30fps video output in MJPG format
- USB 2.0 interface with USB bus-powered operation
- Compact stereo module measuring 30 × 58 × 16 mm
- HM2131 sensor, 3 µm × 3 µm pixel size
- Optical configuration: F/2.2, 4E + IR lens construction, and 75° diagonal field of view (D75°)
- Optical wavelength specifications: 650 ± 10 nm and 850 ± 10 nm
- SMT assembly and anti-static tray packaging

3. Technical Specifications

Item	Specification
Model	TS2M3189
Module Type	Stereo Camera Module
Image Sensor	HM2131
Sensor Size	1/2.7 inch (supplier-stated)
Sensor Type	CMOS; shutter type not specified
Resolution	1920 × 1080
Pixel Size	3 μm × 3 μm
Video Output	MJPG / YUV
Maximum Frame Rate	1080P @ 30fps (MJPG)
Lens Type	650 nm + 850 nm (supplier-stated optical configuration)
Lens Construction	4E + IR
Aperture	F/2.2
Back Focal Length (BFL)	3.8 mm
Field of View	75° diagonal (D75°)
IR Filter	650 ± 10 nm + 850 ± 10 nm
Focus / Object Distance	Object distance: 60 cm (focus type not specified)
WDR / HDR	Not specified
Interface	USB 2.0
Connector	Type / pin count / pitch not specified
Transfer Rate	Not specified; actual throughput depends on the USB implementation and video format
Operating Voltage	5 V DC
Power Supply	USB bus power
Operating Current	Not specified
Operating Temperature	-20°C to +75°C
Stable Imaging Temperature	-15°C to +70°C

4. Mechanical Dimensions



Unit: mm. Dimensions are for reference. Please refer to the latest approved mechanical drawing for production.

Module Dimensions: 30 × 58 × 16 mm

5. Notes

- Specifications are subject to change without prior notice.
- Actual imaging performance may vary depending on lens configuration, lighting conditions, operating temperature, and host platform.
- The specified operating temperature range is -20°C to +75°C. For stable imaging performance, the recommended temperature range is -15°C to +70°C.
- The specified video output formats are MJPG and YUV. Maximum frame rate and supported pixel formats may vary depending on the selected output format and system configuration.
- The specified object distance is 60 cm. Please confirm the required focusing range for your application.
- Optical wavelength specifications are 650 ± 10 nm and 850 ± 10 nm. Please contact TUXIVISION to confirm the optical configuration for your application.
- Connector pin assignments, operating current, storage temperature, and detailed mechanical drawings should be confirmed before final hardware integration.

- Operating system compatibility should be verified against the target hardware and software environment.
- RoHS compliance documentation is available upon confirmation.
- Please contact TUXIVISION for the latest technical specifications, mechanical drawings, interface definitions, and customization requirements.